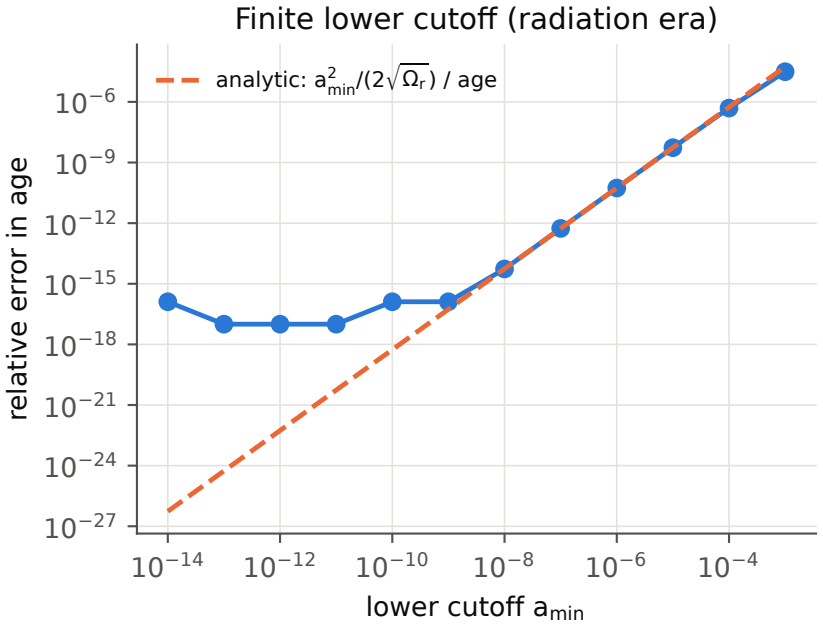
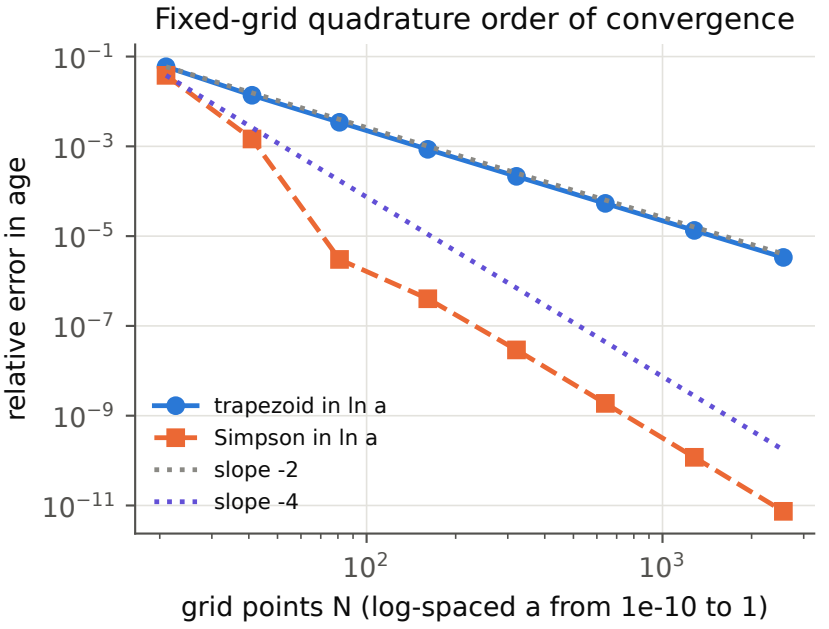
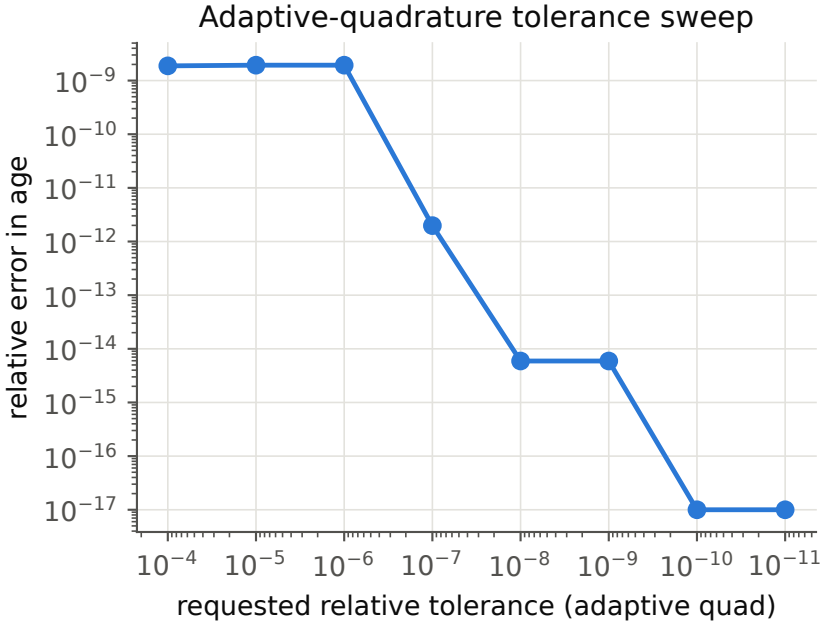
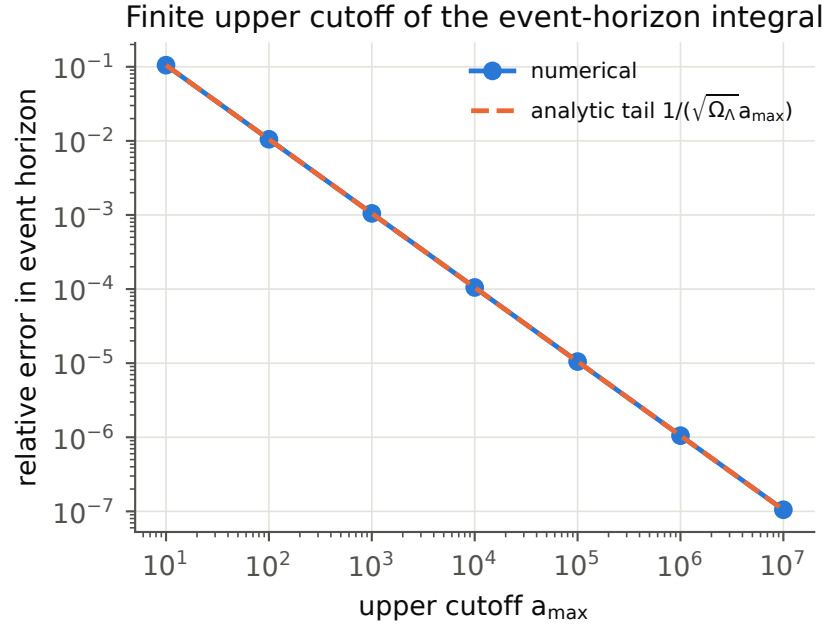

Finite lower cutoff (radiation era)
analytic: $a_{min}^2/(2\sqrt{\Omega_r})$ / age
relative error in age
lower cutoff a_{min}


Fixed-grid quadrature order of convergence
trapezoid in ln a
Simpson in ln a
slope -2
slope -4
relative error in age
grid points N (log-spaced a from 1e-10 to 1)


Adaptive-quadrature tolerance sweep
relative error in age
requested relative tolerance (adaptive quad)


Finite upper cutoff of the event-horizon integral
numerical
analytic tail $1/(\sqrt{\Omega_\Lambda}\,a_{max})$
relative error in event horizon
upper cutoff a_{max}